

Assistive electronic devices

There are a number of assistive electronic devices or rehabilitation equipment to aid people with injuries or physical disabilities. These are tools, products or types of equipment that help people with disabilities or injuries, or the elderly perform daily tasks and activities. These devices may help to move around, see, communicate, eat or get dressed/undressed, and help a person improve the quality of life and maintain a sense of independence.

Well-designed high-quality assistive devices promote greater independence by enabling people to perform tasks that they were formerly unable to accomplish or had great difficulty accomplishing, by providing enhancements to or changed methods of interacting with the technology needed to accomplish such tasks.

An assistive device could be a wheelchair, smartcane for the blind, hearing aid, Braille smartphone, smartbelt, smarting or a disability product (like a bionic organ). Certain devices like eyeglasses and hearing aids require an expert's assessment, but devices like wheelchairs, walkers, bath seats and grab bars are easily obtainable.

Given below are some high-tech assistive electronic devices that are helping those with disabilities lead full and fulfilling lives.

Assistive devices for mobility/ambulation can also be referred to as ambulatory aids. Such aids like canes, crutches and walkers provide an extension of the upper extremities to help transmit body weight and provide support for the user.

A combination of electronics and computer devices gives individuals suffering from paralysis, cerebral palsy and stroke the ability to participate in spoken communication using only their eyes. Using a sophisticated eye-tracking system, they can interact with an onscreen keyboard, allowing them to enter

words and phrases, which are then translated into spoken text via the device's text-to-speech mechanism.

SmartCane is an electronic travel aid that fits on the top-fold of the white cane. It serves as an enhancement to a regular white cane and overcomes its limitations by detecting knee-above and hanging obstacles. For safe mobility, it is important that such obstacles are detected early.

The cane has other uses as well. As a spatial-awareness device, it can detect the presence of objects in its surroundings. Also, as compared to



Hearing aid

the white cane, the detection distance is increased from 0.5 meters to three meters. It informs about the presence of objects before actually touching these with the cane and thus helps in preventing unwanted contact. This avoids socially-awkward situations like collision with people while walking or unsafe collision with animals or into trash.

SmartCane uses ultrasonic range to detect objects in its path and generates tactile output in the form of different vibratory patterns. These vibrations convey the distance information and thus enable the user to negotiate obstacles from a safe distance. With simple orientation and training, any visually-impaired person who is a regular user of the white cane from mobility can benefit from this device.

Travelling alone can be a challenge for the visually impaired, whether it is across the country or down the street. There is always the possibility of taking a wrong turn

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